Arista Access Node MAR-ACC-04: QoS Congestion Event Resolution SOP (ALARM_5209)

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5209

Description

The QoS congestion event (microbursts, interface tail drops, QoS queue starvation) on Arista Access Node MAR-ACC-04, located at Marathahalli Core, is a critical failure domain that necessitates prompt resolution. This SOP outlines the diagnostic and resolution procedures for this event, leveraging the Arista CLI dialect. The root cause of this failure domain lies in the intricate interplay between the QoS policy configuration, interface buffer sizes, and the underlying hardware architecture of the Arista 7280SR. Specifically, the Arista 7280SR's interface buffer sizes (controlled by the \"buffer\" command) and QoS policy configuration (managed via the \"class-map\", \"policy-map\", and \"interface\" commands) interact to determine the interface's capacity for handling microbursts and QoS queue starvation. Furthermore, the Arista's hardware architecture, which includes the use of shared buffers across multiple interfaces, can contribute to interface tail drops and QoS queue starvation in the event of microbursts or other high-bandwidth traffic patterns.

Pre-requisite Checks

- check: Verify that the redundant link at Marathahalli Core is operational and capable of handling the traffic load.

- check: Ensure that all remote hands personnel dispatched to Marathahalli Core are equipped with the necessary tools and knowledge to perform the diagnostic and resolution procedures outlined in this SOP.

- check: Confirm that all necessary permissions and access rights have been granted to the personnel performing the diagnostic and resolution procedures.

Potential Root Causes

Cause	Probability
Insufficient buffer size allocation on the affected interface	High
Misconfigured QoS policy affecting the affected interface	Medium
Hardware limitations (shared buffers) contributing to interface tail drops and QoS queue starvation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status | grep MAR-ACC-04, Expected Output: MAR-ACC-04 \(up, 10G\), Description: This command verifies the operational status and speed of the interface on Node MAR-ACC-04.

Command: show queue interface MAR-ACC-04 | grep buffer, Expected Output: buffer 4096 bytes (default), Description: This command
```

```
displays the buffer size allocation on the affected interface,
providing insight into potential buffer size issues.

Command: show policy-map interface MAR-ACC-04, Expected Output: Policy
Map 'MAR-ACC-04-policy-map', Description: This command displays the
QoS policy configuration applied to the affected interface, helping to
identify potential policy-related issues.
```

Resolution Steps

Step 1: Step: Adjust Buffer Size Allocation, Command: buffer 8192 bytes, Description: This command adjusts the buffer size allocation on the affected interface to a larger value, mitigating potential buffer size-related issues.

Step 2: Step: Review and Revise QoS Policy Configuration, Command: config policy-map MAR-ACC-04-policy-map, Description: This command prompts the user to review and revise the QoS policy configuration applied to the affected interface, ensuring that it is optimized for the specific traffic patterns and requirements of the network.

Step 3: Step: Apply Revised QoS Policy Configuration, Command: apply policy-map MAR-ACC-04-policy-map, Description: This command applies the revised QoS policy configuration to the affected interface, ensuring that it is properly configured to handle the traffic load.

Post-Check Verification

```
Command: show interface status | grep MAR-ACC-04, Expected Output:
MAR-ACC-04 \(up, 10G\), Description: This command verifies the
operational status and speed of the interface on Node MAR-ACC-04 after
the resolution procedures have been completed.

Command: show queue interface MAR-ACC-04 | grep buffer, Expected
Output: buffer 8192 bytes, Description: This command displays the
buffer size allocation on the affected interface after the resolution
procedures have been completed, ensuring that it has been adjusted to
a larger value.

Command: show policy-map interface MAR-ACC-04, Expected Output: Policy
Map 'MAR-ACC-04-policy-map' (revised), Description: This command
displays the revised QoS policy configuration applied to the affected
interface, ensuring that it has been properly optimized for the
specific traffic patterns and requirements of the network.
```

Roll-back Procedures

- Step: Restore Original Buffer Size Allocation, Command: buffer 4096 bytes, Description: This command restores the original buffer size allocation on the affected interface, reverting any changes made during the resolution procedures.

- Step: Restore Original QoS Policy Configuration, Command: rollback policy-map MAR-ACC-04-policy-map, Description: This command restores the original QoS policy configuration applied to the affected interface, reverting any changes made during the resolution procedures.